

Data Visualization using Tableau

Session 1

Agenda

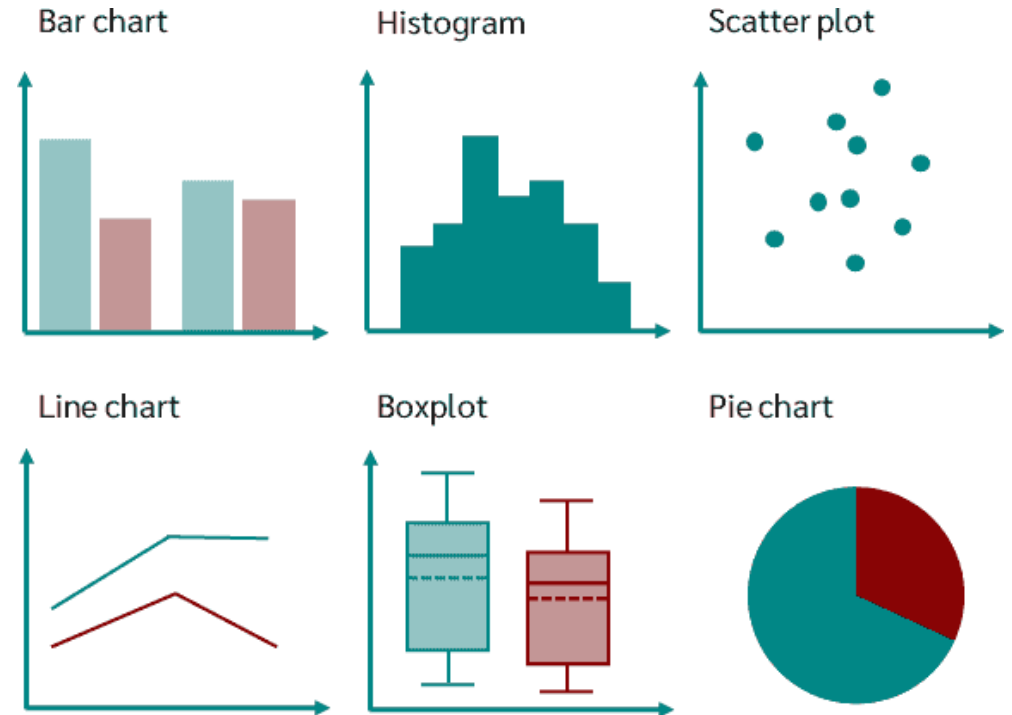
- Define charts and dashboards
- What is Tableau?
- Tableau Installation and Setup
- Tableau Desktop vs Tableau Desktop Public
- Get Started
 - Connect to your data
 - Tableau workspace controls and elements
 - Drag and drop
 - Focus your results
 - Explore your data geographically
- Let's practice

Introduction to Data Visualization

- **Definition:** Transforming data into a graphical or pictorial format to facilitate better understanding and decision-making.
- **Significance:** Enhances the ability to quickly identify trends, patterns, and outliers within large data sets.

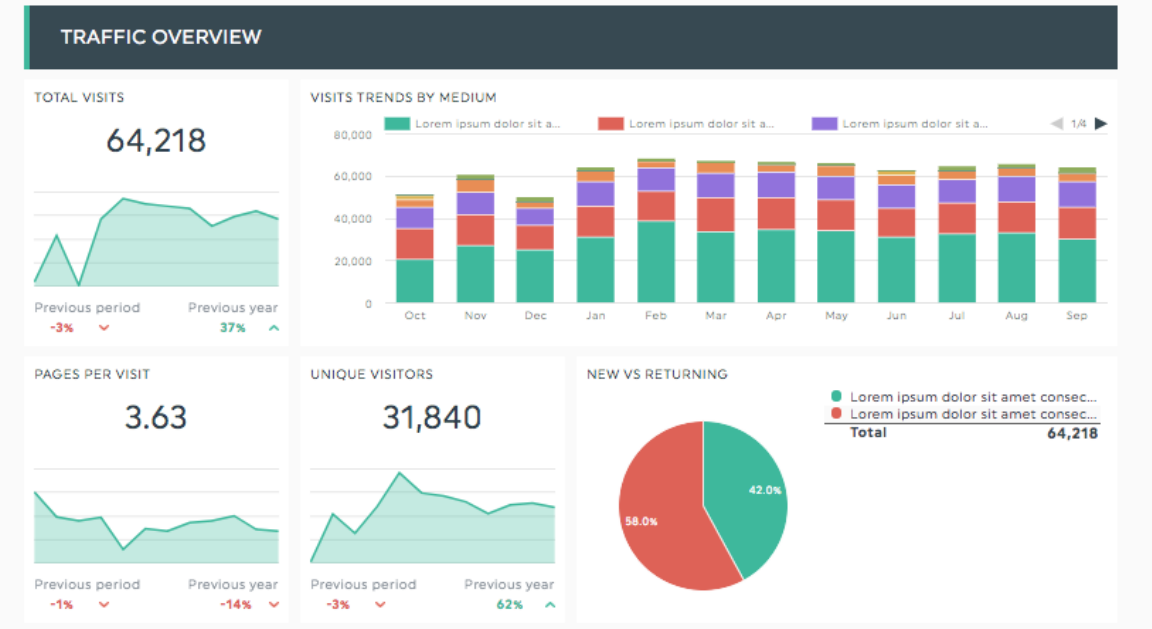
What are Charts?

- **Definition:** Visual representations of data, designed to present information concisely and clearly through various forms such as bars, lines, or circles.
- **Usage:** To simplify the complexity of data and make it understandable at a glance.
- **Choosing the Right Chart:** Depends on the data type, the story you want to tell, and the questions you are answering.



What are Dashboards?

- **Definition:** Information management tools that visually track, analyze, and display key performance indicators, metrics, and data points.
- **Purpose:** To provide an at-a-glance view of business performance to aid in quick decision-making.
- **Usage:** Common in management to aggregate and visualize data from multiple sources.

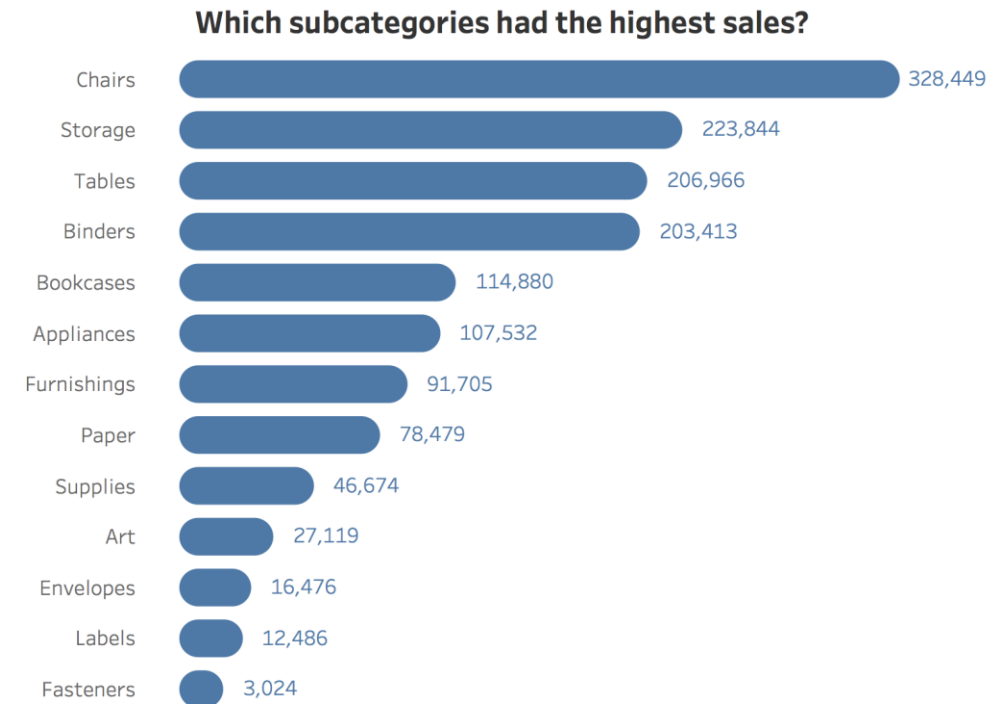


Comparing Charts and Dashboards

- **Charts:** Focus on specific data insights, suitable for detailed data trends and relationships analysis.
- **Dashboards:** Compile multiple charts and data visualizations to offer an integrated view of the data and broader insights.
- **Example:** A single chart might show the monthly sales totals, while a dashboard could display sales totals alongside top products, customer demographics, and regional sales performance to provide comprehensive business insights.

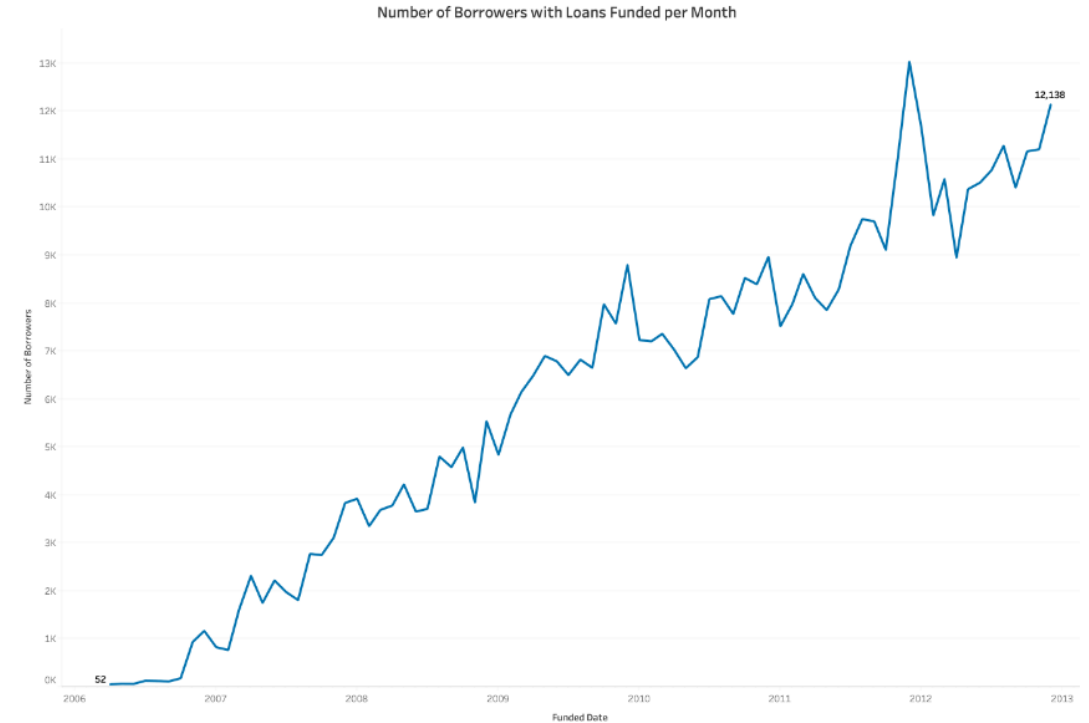
Bar Charts

- **Description:** Used to compare quantities across different categories.
- **When to Use:**
 1. Comparing data across categories.
 2. Showing changes over time (when categories are dates).
- **Example:** Sales performance comparison of different product categories.



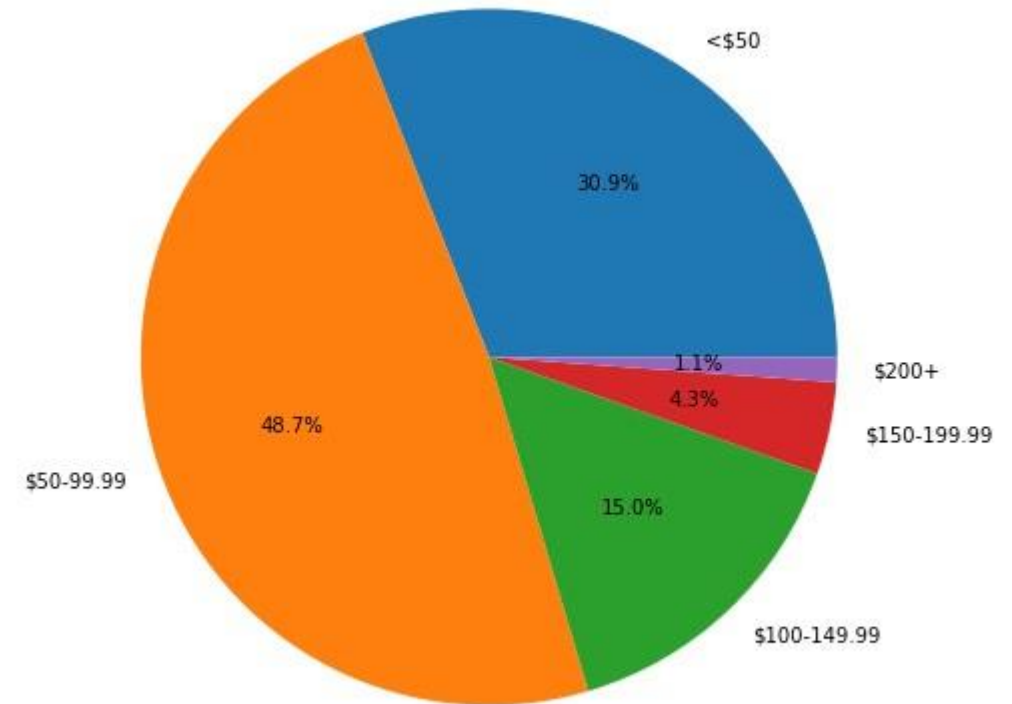
Line Charts

- **Description:** Best for displaying trends over a period of time.
- **When to Use:**
 1. Analyzing trends in data at equal intervals.
 2. Highlighting changes and trends over time.
- **Example:** Analyzing website traffic growth month-over-month.



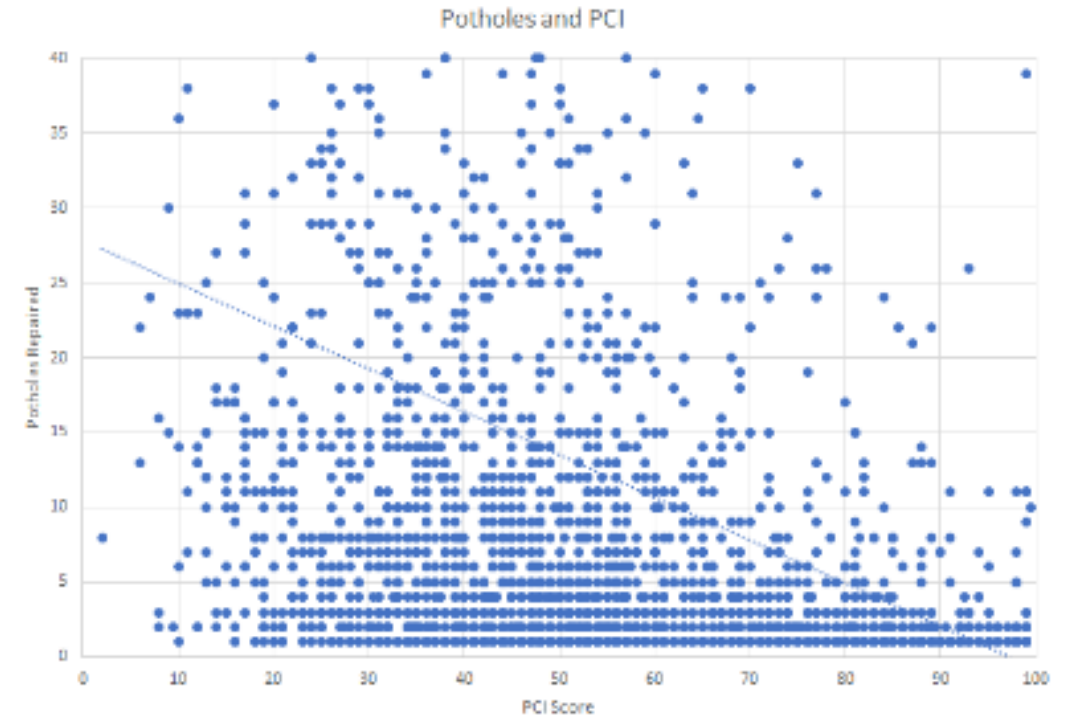
Pie Charts

- **Description:** Show proportions of a whole. Best when categories are few and distinct.
- **When to Use:**
 1. Displaying relative percentages or proportions of a few categories.
 2. Simple distributions which do not require granular analysis.
- **Example:** Market share of different companies in a sector.



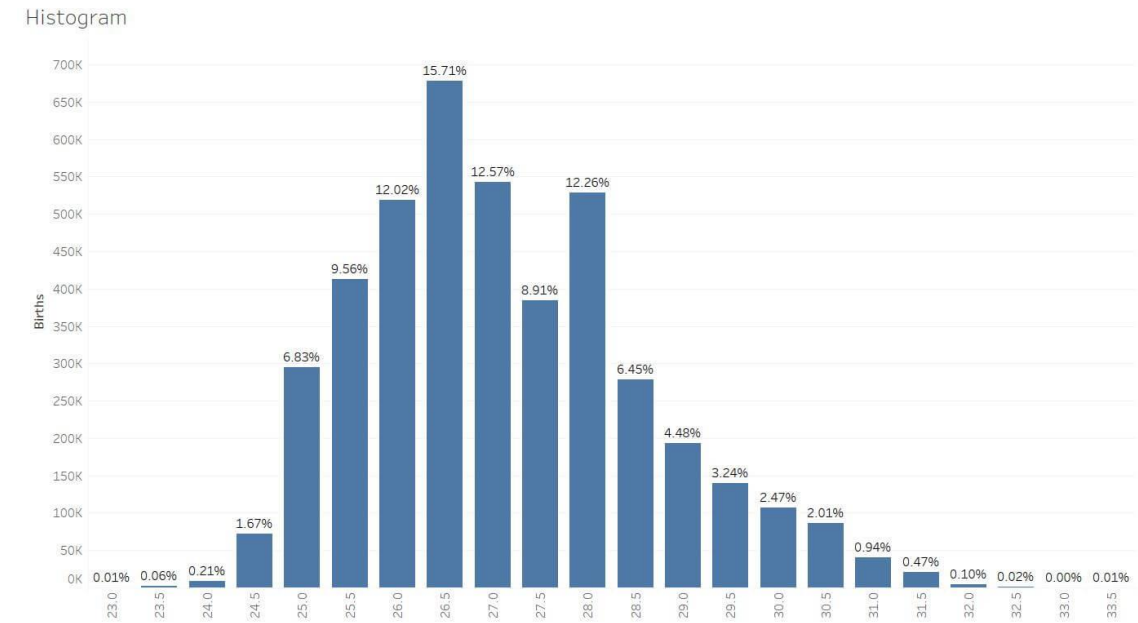
Scatter Plots

- **Description:** Used to display relationships between two variables.
- **When to Use:**
 1. Identifying correlational patterns.
 2. Observing and assessing relationships between variables.
- **Example:** Relationship between advertising spend and revenue generation.



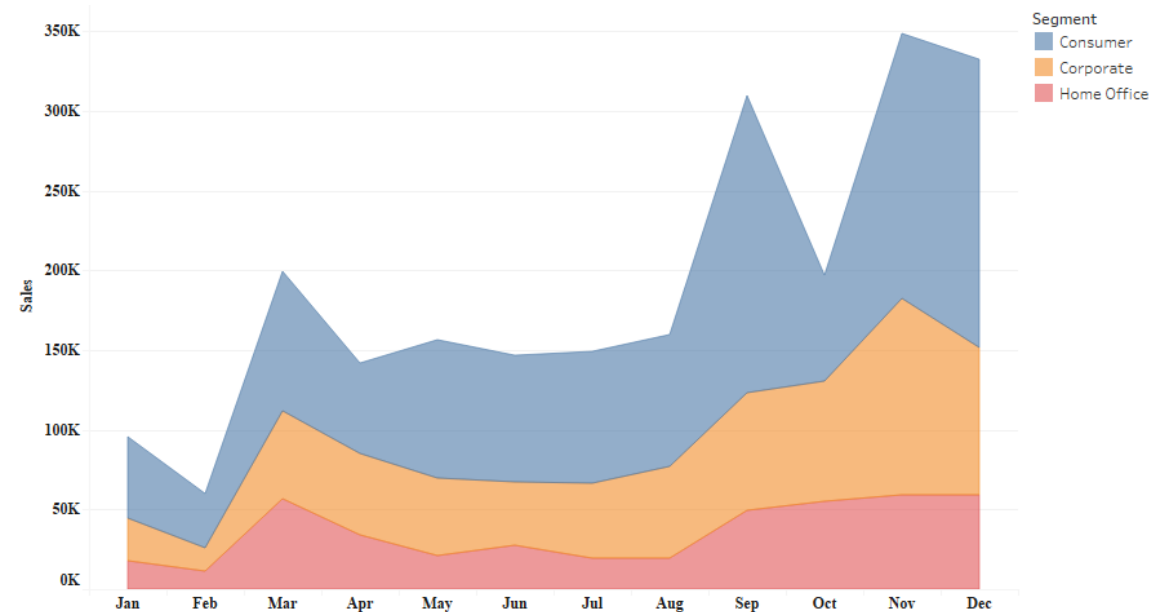
Histograms

- **Description:** Visualize the distribution of data over a continuous interval or certain period.
- **When to Use:**
 1. Understanding the distribution of data.
 2. Identifying clusters and outliers within the data.
- **Example:** Distribution of customer age in a demographic study.



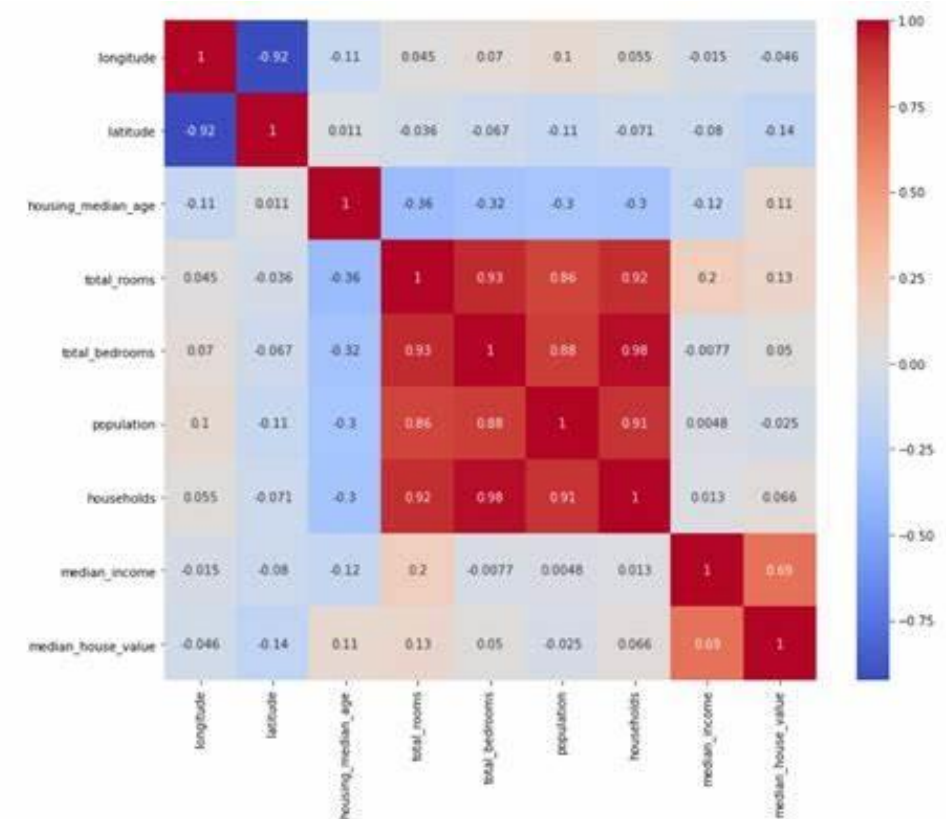
Area Charts

- **Description:** Similar to line charts but filled below the line, emphasizing volume.
- **When to Use:**
 1. Showing the trend over time and also giving a sense of volume.
 2. When you need to highlight the magnitude of change over time.
- **Example:** Total revenue generated over a fiscal year.



Heat Maps

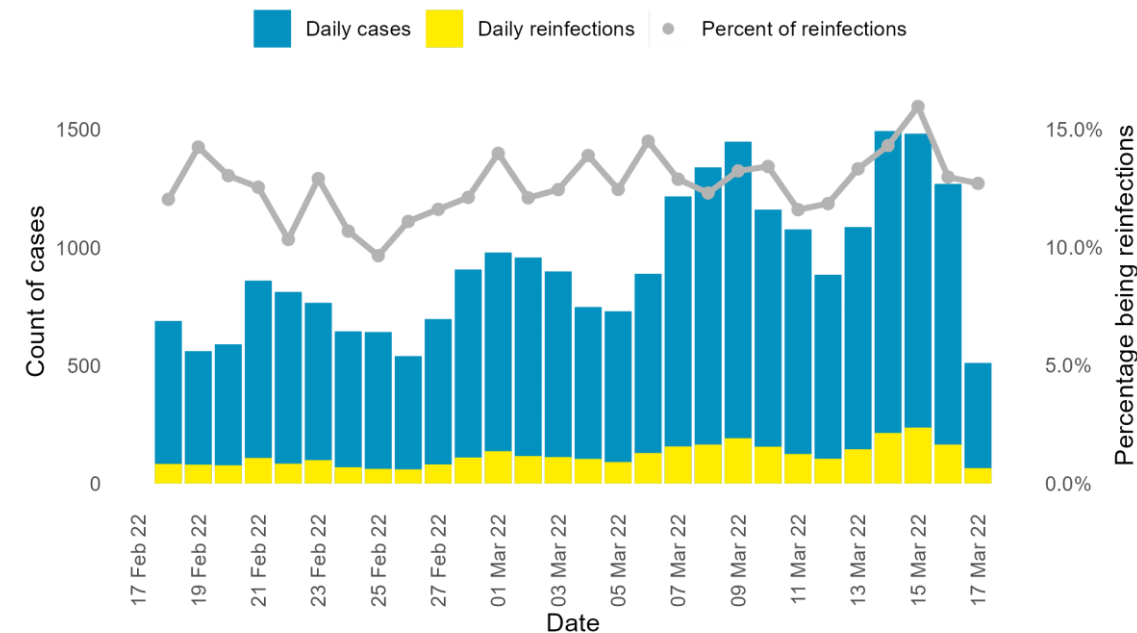
- **Description:** Color coding cells in a matrix according to data values; good for cross-examining multivariate data.
- **When to Use:**
 1. To compare data across two categories.
 2. Highlighting variations influenced by two different factors.
- **Example:** Sales performance analysis across different regions and time periods.



Dual-Axis Charts

- **Description:** Combines two charts to compare different but related data sets with different scales.
- **When to Use:**
 1. When you need to compare two variables that have different scales and are related to each other.
 2. Useful for finding correlations between different variables.
- **Example:** Comparing sales figures with the average temperature in a location.

28 Day Cases by episode and reinfection rate



Source: Public Health Scotland

What is Tableau?

Tableau Software is an interactive data visualization software company focused on business intelligence.

In short, it's an analytics platform that makes it easier for people to **explore** and **manage** data, and faster to discover insights to **make business decisions accordingly**.



Tableau installation and setup

- Go to Tableau Public Product Download
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Connect to the data that matters most to you with Tableau's intuitive drag-and-drop interface to effortlessly create interactive graphs, stunning maps, and live dashboards in minutes. Save your work to your local desktop or publish your viz to your Tableau Public profile. Anyone can do it, it's that easy—and it's free.

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Tableau Desktop vs Tableau Desktop Public

Features and Capabilities	Tableau Desktop Public Edition (free)	Tableau Desktop
Privacy and security	✓	✓
Save visualizations locally	✓	✓
Autorecovery	✓	✓
Unlimited storage	✓	✓
Publish or embed visualizations on other public sites	✓	✓
Download or copy published workbooks	✓	✓
Publish visualizations to Tableau Public	✓	✓
Infrastructure deployment	✓ (Infrastructure available at no cost)	Self-hosted on premise (Role-based licenses)
Share visualizations to Tableau Cloud or Tableau Server	✗	✓
Live data refresh	Limited data refresh (Google Sheets only. Data is refreshed automatically once every 24 hours)	✓
Data sources	Limited data source options (Google Sheets, JSON files, Microsoft Excel 2007 or later, OData, PDF, Spatial files, Statistical files, Text files, and more with Web Data Connector)	✓ (Can connect to all data sources)
Data source row limit	15M rows	Unlimited
Monitoring and metrics	Limited monitoring and metrics	✓

Connect to your data

1. **Tableau icon.** Click  in the upper left corner of any page to toggle between the start page and the authoring workspace.
2. **Connect pane.** Under **Connect**, you can:
 - Connect to data that is stored in a file, such as Microsoft Excel, PDF, Spatial files, and more.
 - Connect to data that is stored on Tableau Server, Microsoft SQL Server, Google Analytics, or another server.
 - Connect to a data source that you've connected to before.
3. Under **Open**, you can open workbooks that you've already created.

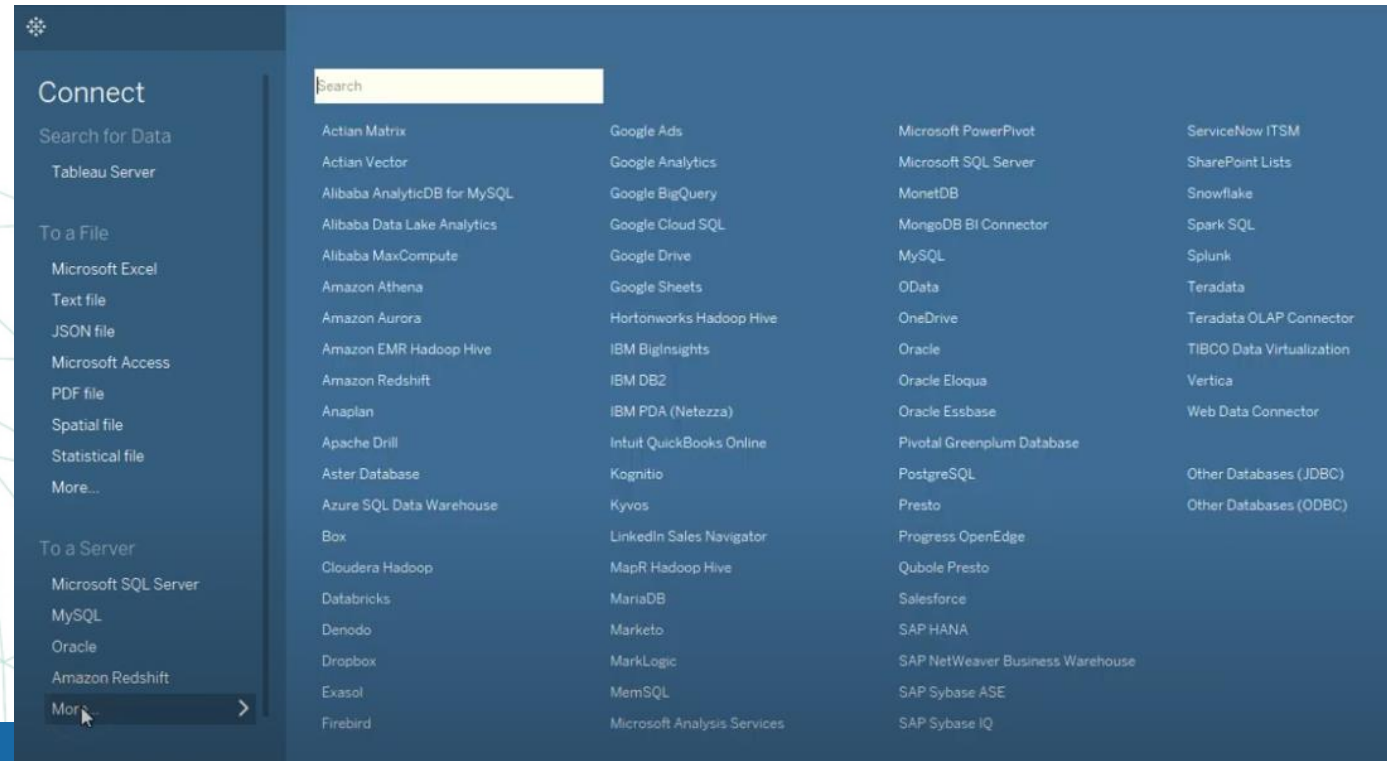


Tableau workspace controls and elements

1. A – Undo
2. B – Data pane
3. C, D – Cards and Shelves
4. E - Show/Hide Cards

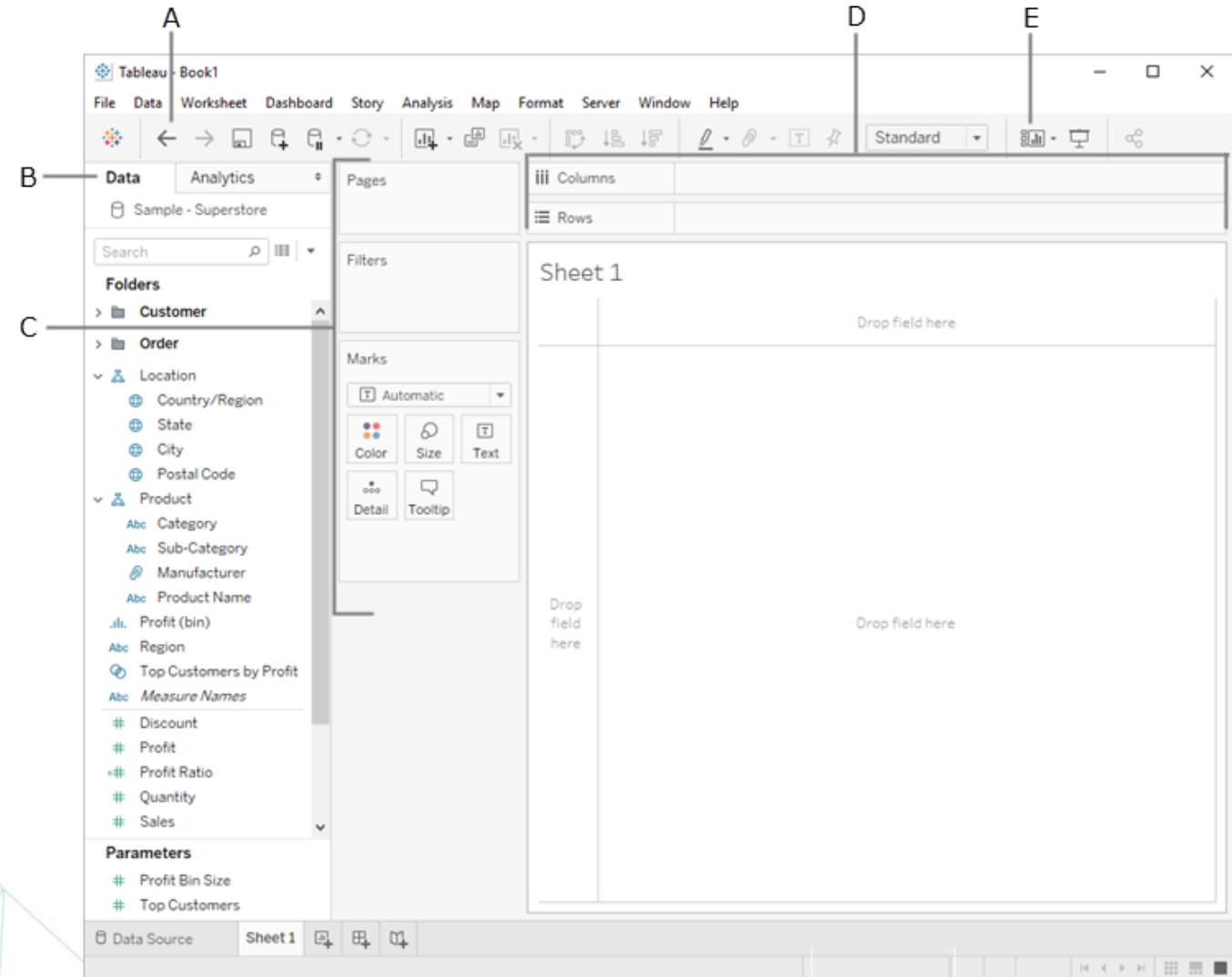
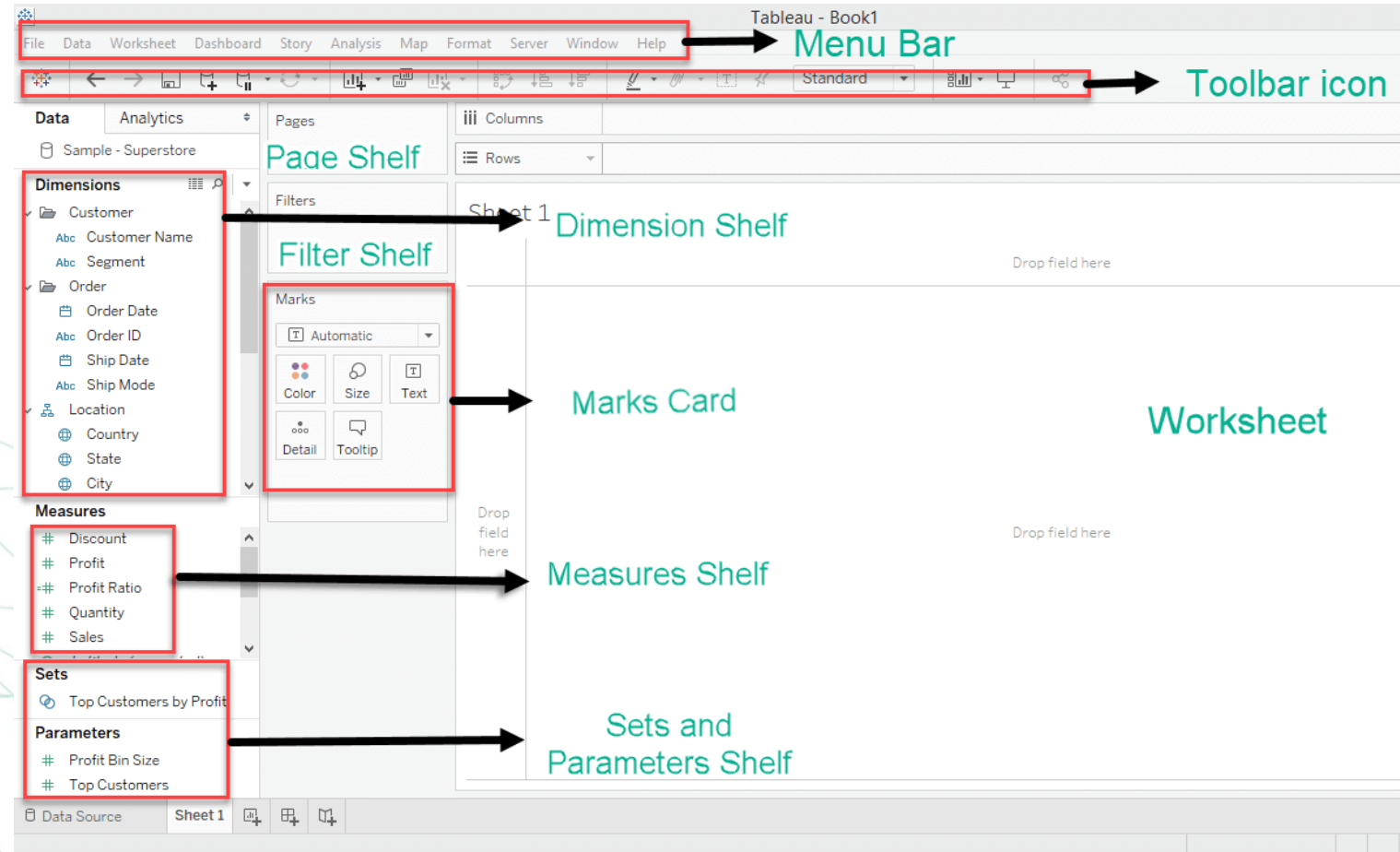
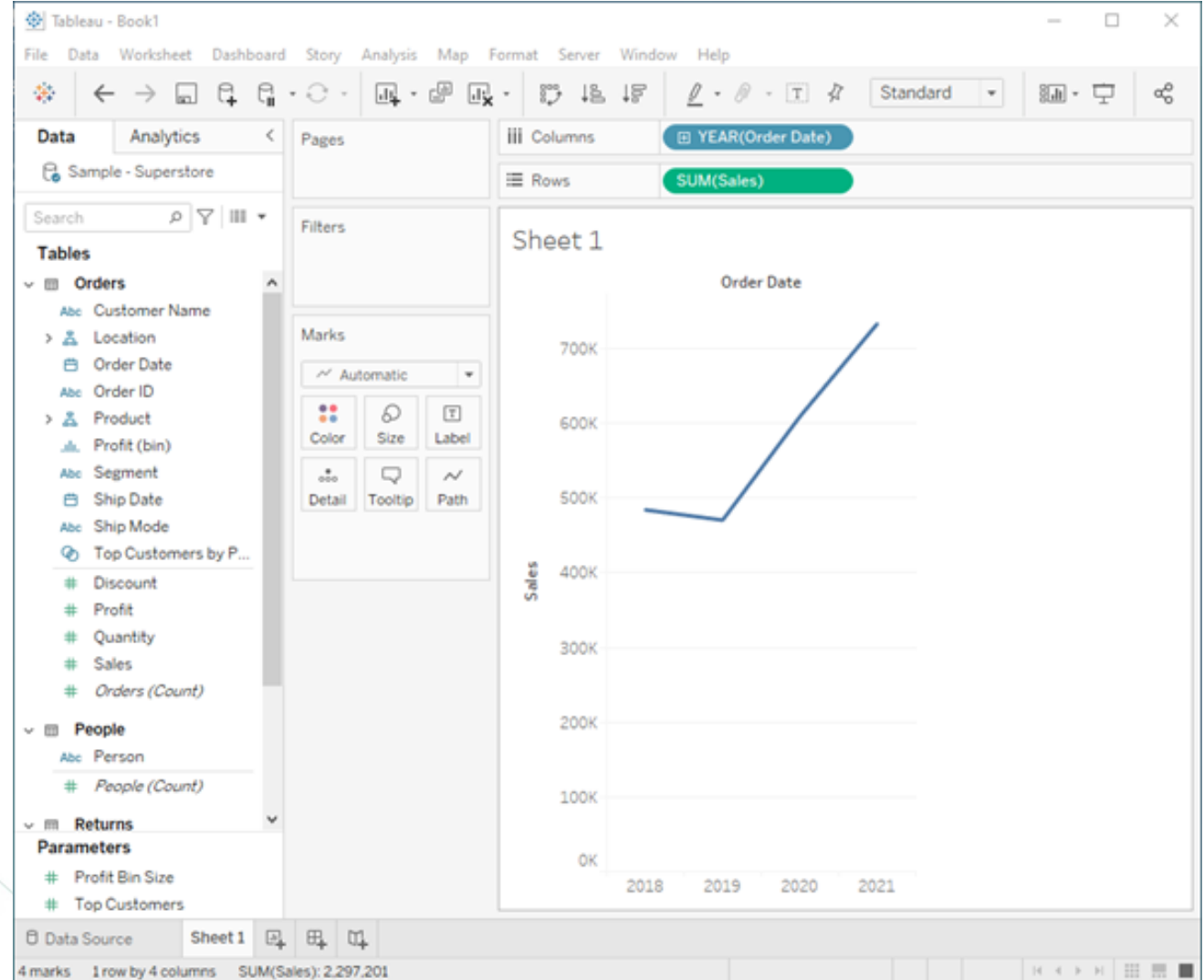


Tableau workspace controls and elements



Drag and Drop

1. From the **Data** pane, drag **Order Date** to the **Columns** shelf.
2. From the **Data** pane, drag **Sales** to the **Rows** shelf.

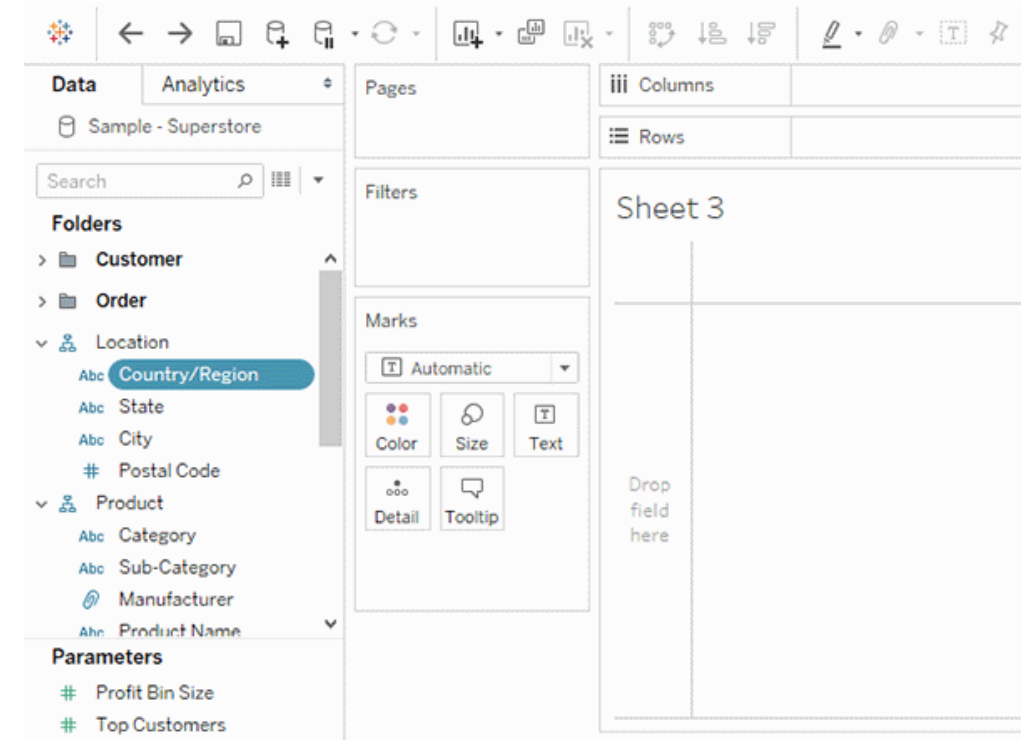


Explore your data geographically

- Geographic fields have globe icons Globe icon that identifies geographic data fields next to them in the Data pane. If a field is assigned a geographic role, Tableau automatically creates a map view when you add the field to Detail on the Marks card.

There are several geographic roles to choose from:

- Airport
- Area Code
- CBSA/MSA
- City
- Congressional District
- Country/Region
- County
- NUTS Europe
- State/Province
- Zip Code/Postcode



Creating Basic Visualizations in Tableau

1. Select the '**Worksheet**' tab to start a new visualization.
2. Drag and drop data fields onto the rows or columns shelves.
3. Choose the type of chart (bar, line, pie) that best represents your data.

Example: Creating a bar chart to compare quarterly sales across different regions.

Customizing Visualizations in Tableau

1. Apply filters to focus on specific data subsets.
2. Adjust color schemes and styles to enhance readability and appeal.
3. Use parameters to make the dashboard interactive.

Example: Adding a filter to a sales dashboard that allows viewers to select a specific year and see data dynamically update.

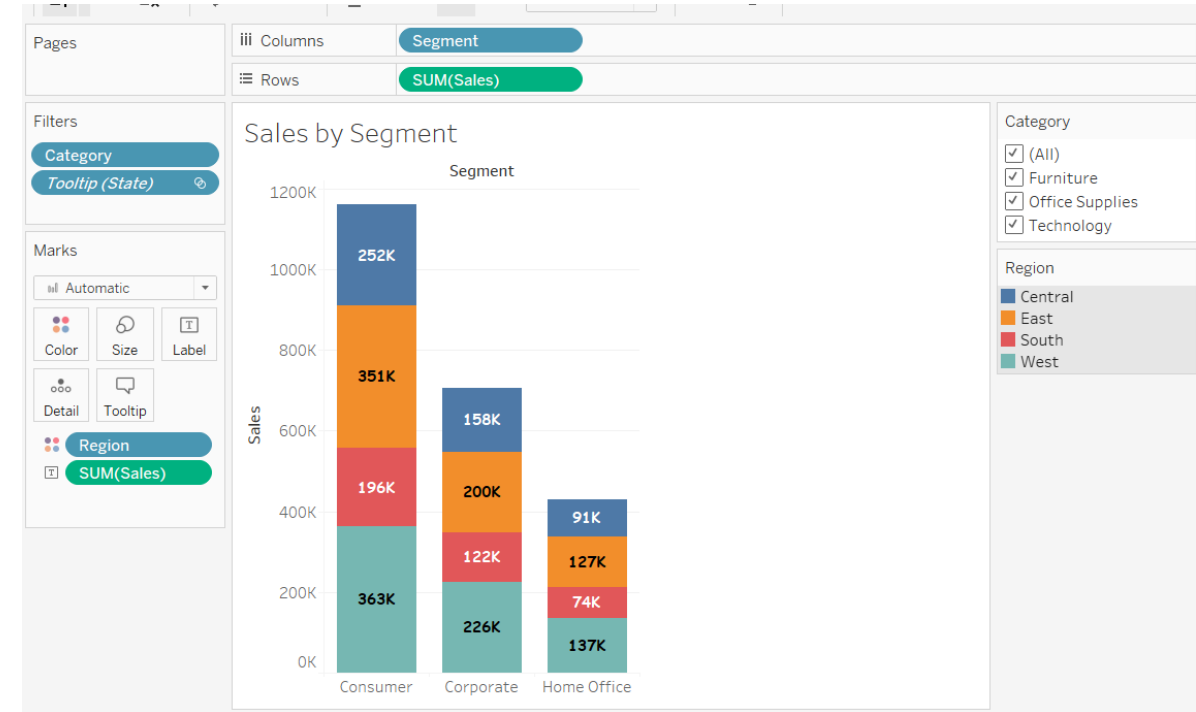
Let's practice

<https://help.tableau.com/current/guides/get-started-tutorial/en-us/get-started-tutorial-connect.htm>



Practicing Making Visuals

- 1- Drag the segment to the columns and the Sales to the rows
- 2- Drag the region to the color
- 3- Drag the sales to the label
- 4- Right click to on the sum(sales) > click Format.
- 5- Change the number to custom and then change the display unit to Thousands.
- 6- Change the font to comic sans and make it bold.



Any Questions ?

Thank You